3.4 Hydroxy Compounds

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.4 Hydroxy Compounds
Difficulty	Medium

Time allowed: 60

Score: /46

Percentage: /100

Question 1a

Alcohol **A** has the formula $(CH_3)_3CCH(OH)CH_3$ and alcohol **B** has the formula $(CH_3)_3CCH_2CH_2OH$

i)

Draw the displayed formula of alcohol **A** and state its name in Table 1.1

[2]

Table 1.1

Name	Displayed formula of alcohol A
………………………………………………	

ii)

In Table 1.2 draw the displayed formula of the organic product formed when alcohol **A** is heated under reflux with acidified potassium dichromate(VI)

[1]

Table 1.2

Displayed formula

iii)

When alcohol **B** is heated under reflux with acidified potassium dichromate(VI), a different organic product is formed.

- Draw the displayed formula of this product in Table 1.3

Table 1.3

Displayed formula of product

- Explain why this product is different to the one formed in part **(ii)**

[4]

[7 marks]

Question 1b

2–bromobutane can be prepared by the following reaction:

$$CH_3CH(OH)CH_2CH_3 + Br^- \rightarrow CH_3CH(Br)CH_2CH_3 + OH^-$$

i)
Name the type of reaction occuring.

[1]

ii)
Suggest suitable reagents for this reaction.

[2]

[3 marks]

Question 1c

Explain why both alcohols **A** and **B** are soluble in water.

[2 marks]

Question 1d

Alcohols **A** and **B** are heated with an alkaline solution of iodine. Complete Table 1.4 stating the observations of for each alcohol.

Table 1.4

Alcohol **A** observation	Alcohol **B** observation

[2 marks]

Question 2a

When pentan-2-ol reacts with concentrated phosphoric acid under reflux, three alkenes are formed, **X**, **Y** and **Z**.

X and **Y** are stereoisomers of one another.

Draw the structures of alkenes, **X, Y** and **Z**.

[3 marks]

Question 2b

State how the stereoisomers formed in part **(a)** arise.

[2 marks]

Question 2c

Pentan-2-ol can undergo different reactions.

Write the equations for the reaction of pentan-2-ol with the following reagents. You do not need to include state symbols

i)
Oxygen

....

[2]

ii)
Sodium

....

[2]

iii)
Phosphorus(V) chloride

....

[2]

[6 marks]

Question 2d

Pentan-2-ol can undergo oxidation when reacted with potassium dichromate(VI) to form a ketone.

Draw and label the apparatus which would be necessary to ensure that pentan-2-ol was oxidised to the ketone.

[4 marks]

Question 3a

This question is about the formation of alcohols.

2-bromobutane reacts with OH^- in aqueous conditions to produce alcohol **A.**

Draw the structure of alcohol **A**

[1 mark]

Question 3b

Draw the reaction mechanism for the formation of alcohol **A** in part **(a)**.

[3 marks]

Question 3c

Alcohol **A** reacts with ethanoic acid in the presence of concentrated sulfuric acid to form compound **B**. Draw the structurs of compound **B**.

[1 mark]

Question 3d

Both alcohol **A** and water can act as weak acids. Explain why alcohol **A** is a weaker acid than water.

[3 marks]

Question 4a

Butan-1-ol can be oxidised by acidified potassium manganate(VII), $KMnO_4$, using two different methods.

In the first method, butan-1-ol is added dropwise to acidified potassium manganate(VII), $KMnO_4$ and the product is distilled off immediately.

Using the symbol [O] for the oxidising agent, write an equation for this oxidation of butan-1-ol, showing clearly the structure of the product.

State what colour change you would observe ....

[1]

Equation ....

[1]

[2 marks]

Question 4b

In a second method, the mixture of butan-1-ol and acidified potassium manganate(VII), $KMnO_4$, is heated under reflux.

Identify the product which is obtained by this reaction.

[1 mark]

Question 4c

Give the structures and names of two branched chain alcohols which are both isomers of butan-1-ol.

Only **Isomer 1** is oxidised when warmed with acidified potassium manganate(VII).

Isomer 1	Isomer 2
Name	**Name** ....
Structure	**Structure**

[4 marks]

Question 4d

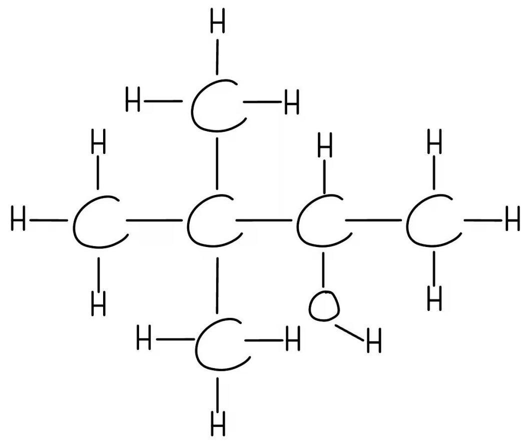
Write the half equation for the reduction of MnO_4^- in this reaction.

[2 marks]

Model Answers: Medium

1a

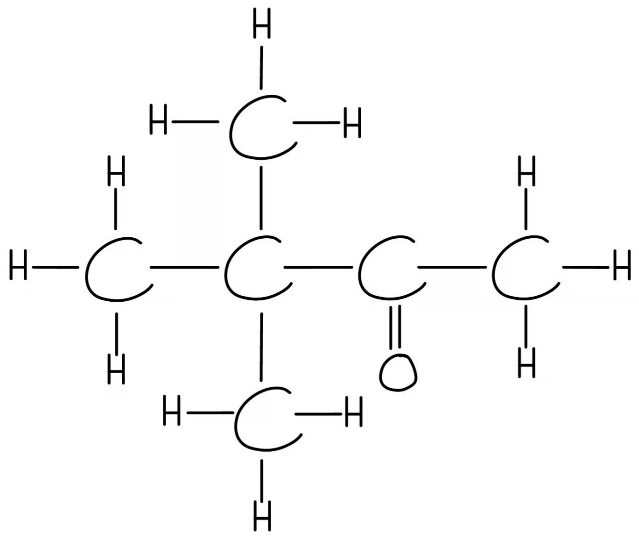
i) The displayed formula and name of alcohol **A** is:

Name	Displayed formula of alcohol A
3,3-dimethylbutan-2-ol	

- Correct displayed formula drawn; [1 mark]
- Correct name stated; [1 mark]

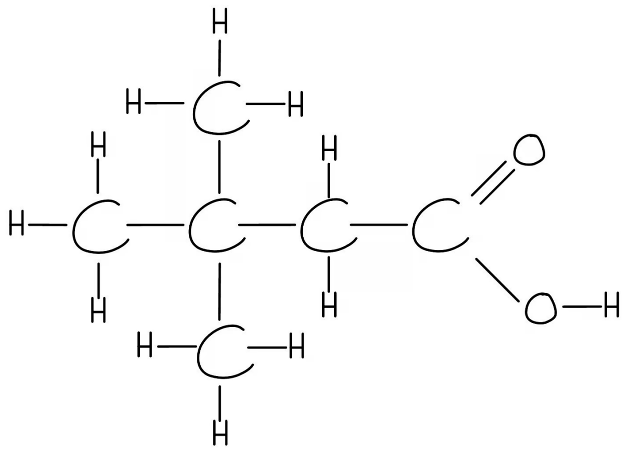
ii) The product formed when alcohol **A** reacts with acidified potassium dichromate(VI) is:

Displayed formula of product



- Correct displayed formula drawn; [1 mark]

iii) When alcohol **B** reacts with acidified potassium dichromate(VI)

Displayed formula of product


- Correct displayed formula drawn; [1 mark]
- Correct name stated; [1 mark]
- A carboxylic acid is formed instead of a ketone; [1 mark]

- As alcohol **B** is a primary alcohol whereas alcohol **A** is a secondary alcohol; [1 mark]

[Total: 7 marks]

- Displayed formulas should show all bonds
- The product formed from the complete oxidation of alcohol **B** is a carboxylic acid as primary alcohols when oxidised will form a aldehyde followed by a carboxylic acid and secondary alcohols will oxidise to a ketone only

1b

b)

i) The name of the type of reaction is:

- Substitution; [1 mark]

ii) Suitable regents are:

- Sodium bromide / NaBr / potassium bromide / KBr; [1 mark]
- Concentrated sulfuric acid / conc H_2SO_4; [1 mark]

[Total: 3 marks]

- The overall reaction for the formation of 2–bromobutane is:
- $CH_3CH(OH)CH_2CH_3 + HBr \rightarrow CH_3CH(Br)CH_2CH_3 + H_2O$
- In order to prepare the bromoalkane the alcohol is treated with a mixture of sodium or potassium bromide and concentrated sulphuric acid
- This produces hydrogen bromide which reacts with the alcohol
- The mixture is warmed to distil off the bromoalkane

1c

c) Alcohols A and B are both soluble in water because:

- They contain O-H bonds / hydroxy groups; [1 mark]
- Therefore hydrogen bonds can be formed with water molecules; [1 mark]

[Total: 2 marks]

- Hydrogen bonding can occur between alcohols as the O atom is very electronegative
- Alcohols interact with water and other alcohol molecules
- **Remember:** It is an O lone pair on one alcohol molecule and the H in the O-H bond of another molecule that interact

1d

d) The observations are:

Alcohol **A** observation	Alcohol **B** observation
Yellow precipitate	Yellow precipitate

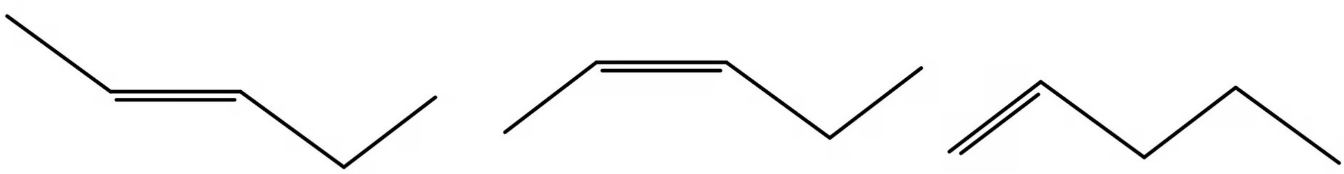
- Correct observation for **A**; [1 mark]
- Correct observation for **B**; [1 mark]

[Total: 2 marks]

- Both alcohols contain the RC-OH group
- When heated with an alkaline solution of iodine, alcohols that contain the RCH_2-OH group will form a **yellow precipitate**
- This is called the iodoform test and is a test for alcohols
  - However methanol does not give a positive result for this test as there is no R group bonded to the carbon
    - CH_3OH

2a

a) The structures of isomers **X**, **Y** and **Z** are:



X **Y** **Z**

Structures X and Y are interchangeable

- Correct structure scores [1 mark] each

[Total: 3 marks]

- The elimination of alcohols will form alkenes, this is also called dehydration as a water molecule is removed
- The elimination of the -OH group in pentan-2-ol will cause a double bond to be formed on either the first or second carbon, forming pent-1-ene or pent-2-ene
- As there are two different R groups either side of the double bond in pent-2-ene, two stereoisomers can be formed
- You are asked to draw the skeletal structure so you will not get the marks if the structural formulas are drawn

2b

b) Stereoisomers are formed by:

- The compounds have a C=C double bond with two different atoms / groups bonded to each carbon; [1 mark]

- The C=C bond cannot rotate; [1 mark]

[Total: 2 marks]

- Pent-2-ene displays stereoisomerism as each carbon atom in the C=C bond has **two different** atoms or groups of atoms attached to it
- The two stereoisomers are *E*-pent-2-ene and *Z*-pent-2-ene

2c

c)

i) The equation with oxygen:

$2CH_3CHOHCH_2CH_2CH_3 + 15O_2 \rightarrow 10CO_2 + 12H_2O$

- Correct reagents and products; [1 mark]
- Correct balancing; [1 mark]

ii) The equation with sodium:

$2CH_3CHOHCH_2CH_2CH_3 + 2Na \rightarrow 2CH_3CHO^-Na^+CH_2CH_2CH_3 + H_2$

- Correct reagents and products; [1 mark]
- Correct balancing; [1 mark]

iii) The equation with phosphorus(V) oxide:

$CH_3CHOHCH_2CH_2CH_3 + PCl_5 \rightarrow CH_3CHClCH_2CH_2CH_3 + HCl + POCl_3$

- Correct reagents and products; [1 mark]
- Correct balancing; [1 mark]

[Total: 6 marks]

- The reaction of an alcohol with oxygen is combustion
- A common error when writing a combustion equation is to balance incorrectly
- It can help when balancing these equations to double the alcohol molecule, this makes it easier to balance the carbon hydrogen and oxygen atoms
    - The order to balance in should be carbons, hydrogens and then oxygen atoms
- You should know the products for the reactions of alcohols with Na, HX (in presence of concentrated acid), PCl_3, PCl_5 and $SOCl_2$

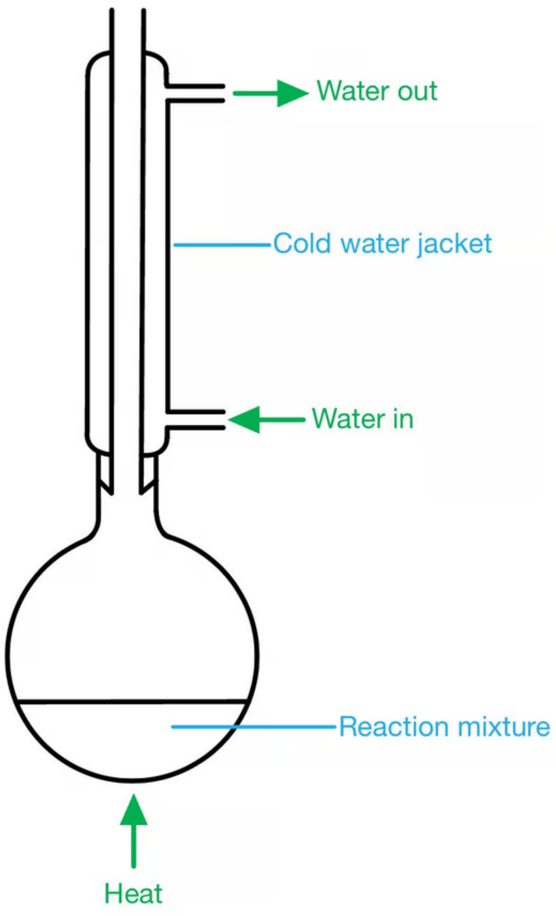
2d

d) The labelled apparatus for the oxidation of pentan-2-ol are:

- The reaction of an alcohol with oxygen is combustion
- A common error when writing a combustion equation is to balance incorrectly
- It can help when balancing these equations to double the alcohol molecule, this makes it easier to balance the carbon hydrogen and oxygen atoms
  - The order to balance in should be carbons, hydrogens and then oxygen atoms
- You should know the products for the reactions of alcohols with Na, HX (in presence of concentrated acid), PCl_3, PCl_5 and $SOCl_2$

2d

d) The labelled apparatus for the oxidation of pentan-2-ol are:



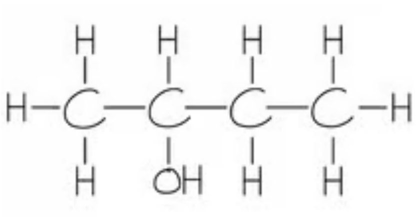
- Heat source labelled; [1 mark]
- Water in and water out correctly labelled on condenser; [1 mark]
- Round or pear shaped flask with condenser set up vertically for reflux; [1 mark]
- No gaps in the apparatus at all, other than the very top where the condenser must be left open and not sealed; [1 mark]

[Total: 4 marks]

- Drawing apparatus is often not done well by students in exams, so make sure you check your diagram for errors
- A common error includes closing the top of the reflux condenser by drawing a line across the top

3a

a) The structure of alcohol **A** is:



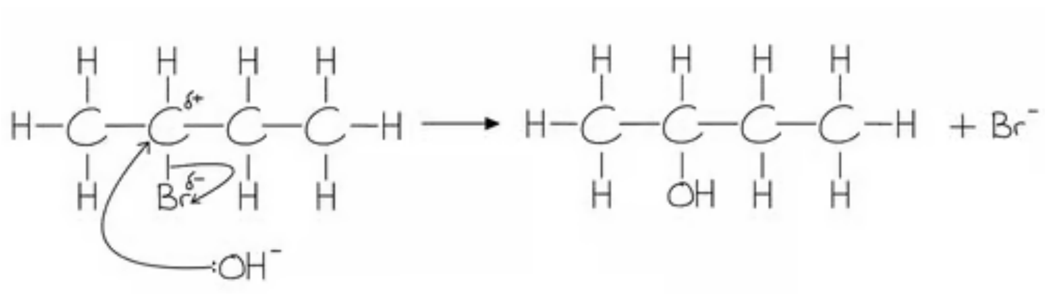
- Correct structure; [1 mark]

[Total: 1 mark]

- The reaction of 2-bromobutane with aqueous alkali results in the formation of an alcohol
- The halogen is replaced by the OH^-
- The aqueous hydroxide (OH^- ion) behaves as a nucleophile by donating a pair of electrons to the carbon atom bonded to the halogen

3b

b) The reaction mechanism for the formation of alcohol **A** in part **(a)** is:



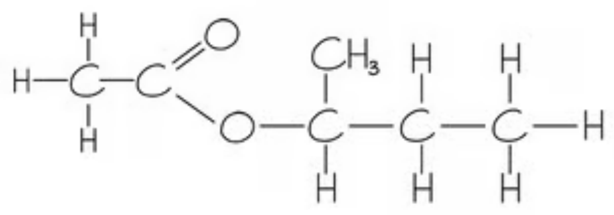
- δ+ carbon and δ- bromine identified; [1 mark]
- Curly arrow from lone pair on OH^- nucleophile to δ+ carbon; [1 mark]
- Curly arrow from C-Br bond to δ- bromine atom; [1 mark]

[Total: 3 marks]

- Make sure in your reaction mechanisms that you have identified the partial charges and your curly arrows start from an area of negative charge and go towards an area of positive charge

3c

c) The structure of compound **B** is:



- Correct structure of the ester; [1 mark]

[Total: 1 mark]

- The secondary alcohol will bond with ethanoic acid by the oxygen atom, so the alkyl chain will be branched

3d

d) Alcohol **A** is a weaker acid because:

- Alcohol A contains R groups / water does not have an R group; [1 mark]
- R group is electron donating so causes a positive induction effect; [1 mark]
- Therefore more electron density on the O^- atom; [1 mark]

[Total: 3 marks]

- When water dissociates, the hydroxide ion only has one other hydrogen atom
  - H_2O (l) ⇌ H^+ (aq) + OH^- (aq)
- There is no extra electron density on the oxygen which is less likely to accept an H^+ ion
- Water is therefore a stronger acid than alcohols
- When an alcohol dissociates an alkoxide ion, R^+ , is formed
  - ROH (aq) ⇌ R^+ (aq) + OH^- (aq)

- The alkoxide ion is more likely to accept an H^+ ion and form the alcohol again, making it a weaker acid

4a

a)

i) The colour change is:

- Purple / pink to colourless; [1 mark]

ii) The equation is:

- $CH_3CH_2CH_2CH_2OH + [O] \rightarrow CH_3CH_2CH_2CHO + H_2O$; [1 mark]

[Total: 2 marks]

- Butan-1-ol is a primary alcohol so is initially oxidised to an aldehyde

4b

ii) The product is:

- Butanoic acid; [1 mark]

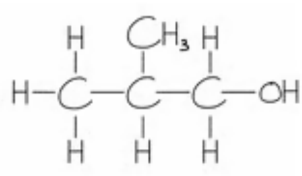
[Total: 1 mark]

- Reflux apparatus will fully oxidise the primary alcohol to a carboxylic acid
- Butan-1-ol will oxidise to butanal and then butanoic acid

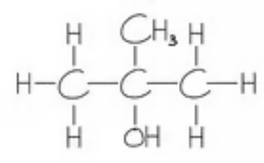
4c

c) The name and structure of the isomers are:

Isomer 1

- 2-methylpropan-1-ol; [1 mark]
-  ; [1 mark]

Isomer 2

- 2–methylpropan-2–ol; [1 mark]
- 

; [1 mark]

[Total: 4 marks]

- Isomers of butan-2–ol must be either a primary or tertiary alcohol
- Isomer 2 must be a tertiary alcohol and therefore can not be oxidised by potassium manganate(VII), only primary and secondary alcohols can be oxidised

4d

d) The balanced half equation is:

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

- Mn^{2+} species identified; [1 mark]
- Correctly balanced half equation; [1 mark]

[Total: 2 marks]

- To balance this half equation you need to
- Write out the ions
    - $MnO_4^- \rightarrow Mn^{2+}$
- Balance the oxygen atoms using water
    - $MnO_4^- \rightarrow Mn^{2+} + 4H_2O$
- Balance the hydrogen atoms using H+
    - $MnO_4^- + 8H^+ + \rightarrow Mn^{2+} + 4H_2O$
- Balance the charge using electrons - the charge is 7+ on the left hand side and 2+ on the right hand side, so 5 electrons are required
    - $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$